

The metabolic regulation of aging

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Here we review the environmental and genetic manipulations that link cellular and organismal metabolism to aging. In particular, we explore how nutrients are sensed and how various intracellular energy nodes seem to coordinate distinct metabolic alterations linked to extended longevity. In addition, the role of mitochondria as both a metabolic and signaling organelle is discussed. Finally, we review a host of new targeted pharmacological approaches that attempt to exploit the connection between aging and metabolism to treat a wide range of age-related diseases. Together, these insights are beginning to reveal answers to century-old mysteries and are providing a future road map for the rational extension of lifespan.

At the turn of the sixteenth century, Italian nobleman Luigi Cornaro found himself near death. Although he was only in his mid-thirties, years of gluttony and excess had taken their toll. Fearing his time was near, he vowed to change his lifestyle and decided to limit his future daily consumption of food to approximately 350 g (mostly bread, egg yolk and soup) accompanied by two glasses of wine per day¹. He would religiously follow this diet for the next six decades until his death at the age of 102. Cornaro published a series of treatises, including *Discorsi della vita sobria* (Discourses on the Sober Life) that described the increased vitality and remarkable longevity that he attributed to his self-imposed dietary restrictions. Though now largely forgotten, these manuscripts provided one of the first tangible links between energy utilization, metabolism and the process of aging.

Cornaro's early ideas have been bolstered by a recent avalanche of more controlled experiments, which have demonstrated in a wide range of organisms that reduction of food intake correlates with an extension in lifespan². Included among this menagerie are various model organisms conducive to genetic manipulation. Research that combined a reduction of energy intake with genetically modified organisms has revealed key pathways that link nutrient sensing with longevity. The identification of these targets has subsequently led to pharmacological interventions that can potentially serve as dietary-restriction mimetics: small molecules that somehow fool the body to act as if nutrients are always scarce. Over the years, investigators seeking to understand the connection between metabolism and aging have placed particular emphasis on understanding how mitochondria might participate in this process. Whole-genome RNAi-based screens in model organisms have furthered this interest, consistently identifying nuclear-encoded mitochondrial proteins as important regulators of longevity^{3,4}. These and other observations, including mouse models of mitochondrial-dependent aging^{5,6}, have tightened the association between bioenergetics and lifespan, although

substantial gaps still remain regarding the nature and importance of these connections.

Here I review the link between metabolism and aging. I place particular emphasis on those longevity pathways that sense nutrient availability, the evolving role that mitochondria might have in the aging process, how an understanding of these energetic pathways might lead to new insights into age-related diseases and the growing indication that, by following in Cornaro's footsteps, we may soon turn aging into a pharmacologically modifiable condition.

Metabolic effects of dietary restriction

Energy partitioning. Aging is accompanied by a host of metabolic changes, including modulation of mitochondrial function, a decline in insulin sensitivity and alterations in substrate utilization⁷. These changes may contribute to the aging phenotype and predispose an individual to age-related conditions, but correlation and causation are not easily distinguished by assessing metabolic differences between young and old organisms. Perhaps more insight can be obtained by dissecting how metabolic pathways are affected by environmental or genetic manipulations that are known to regulate longevity. In mice and rats, lifelong caloric restriction (CR) can, under certain circumstances, extend lifespan by up to 50% compared to that of *ad libitum*-fed control animals⁸. The magnitude of this effect has prompted considerable efforts to understand how limiting food intake translates into longer life and, in particular, the genetic and biochemical pathways mediating this effect.

It is important to realize that even a seemingly simple dietary regime such as CR can produce a myriad of physiological and metabolic alterations that may or may not be relevant to the observed increase in lifespan (Fig. 1). For instance, when CR is begun early in life, mice placed on a CR diet are smaller than animals that are fed a control diet⁸. This has led to considerable speculation that the lifespan benefits of CR have some tradeoffs between the growth of the organism and ultimate longevity. This concept of energy partitioning is crucial for understanding how metabolism might regulate the aging process. Nonetheless, the relationship between organismal size and longevity is complex. For instance, the first long-lived, genetically modified mouse model was the Ames dwarf mouse⁹. These animals

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Received 19 June; accepted 26 October; published online 8 December 2015; doi:10.1038/nm.3998

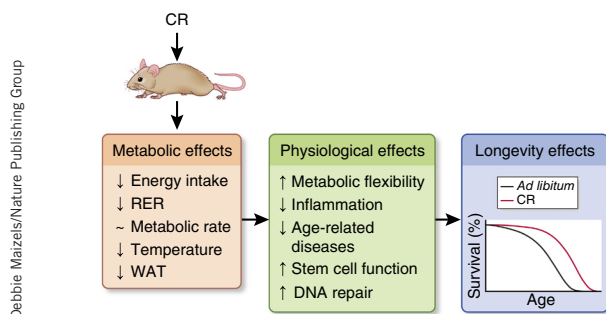


Figure 1 The myriad effects of caloric restriction (CR). Reduction of nutrient intake results in many metabolic and physiological changes. The net effect of these changes is an increase in the median and maximum lifespan of a wide array of organisms. RER, respiratory exchange ratio; WAT, white adipose tissue.

have a mutation in the paired-like homeodomain factor 1 (*Prop1*) gene that leads to a marked alteration in the function of the anterior pituitary gland, which in turn leads to a reduction both in growth hormone release from the pituitary and in insulin and insulin growth factor 1 (IGF1). These factors all contribute to the reduced size of the Ames dwarf mouse. Whereas Ames mice and mice that have undergone CR are similar in size and share similar increases in overall lifespan, evidence suggests that the mechanism that enables the small Ames mice to live longer may be mechanistically distinct from the lifespan extension seen after nutrient restriction. First, when normalized to body size, food intake for Ames mice actually exceeds that of control mice¹⁰. More importantly, when Ames mice are placed on a CR diet, they can experience further extended lifespans, with median lifespans that are nearly twice those of control mice fed a normal diet¹¹. Thus, although CR can result in a smaller animal, and although many genetic models that extend lifespan have similar effects on body size, the mechanisms that underlie the observed longevity benefits in these various models do not seem to completely overlap.

Species-specific effects of caloric restriction. The metabolic effects of CR can be profound but species specific. In small rodents, a several-degree difference in body temperature is observed⁸. In contrast, in a healthy human population when individuals are randomized to consume either a normal diet or a diet intended to reduce caloric intake by 25%, core body temperatures are reduced very modestly (0.2 °C)¹² if at all¹³. In both experimental mouse models and humans, CR does have marked effects on weight and white adipose tissue (WAT) biology. Although overall weight loss is common, a large fraction of the weight loss comes from the WAT and, in particular, visceral fat stores^{14,15}. Interestingly, the common strains of laboratory rodents all experience extended lifespans in response to CR, but this is not a universal effect. Indeed, certain inbred mice strains show only a modest effect or even a reductive effect on lifespan after CR¹⁶. One variable that seems to correlate with the beneficial effects of CR is the degree of fat loss induced by food restriction¹⁷. Paradoxically, in an analysis of nearly 40 strains of mice, the beneficial longevity effects of CR were actually correlated with those mice that were best able to maintain fat stores, in the setting of a 40% CR regimen¹⁷.

Whereas the effects of CR on body temperature and body fat are quite consistent and reproducible, the effect of CR on basal metabolic rate (BMR) is more of an unsettled question. In general, resting metabolic rate is determined by oxygen consumption normalized to body size. By using such strategies, a number of investigators have

concluded that BMR does not change in the setting of CR, whereas other studies have led to the conclusion that BMR is reduced⁸. These discrepancies are in part due to differences in how the measurement of oxygen consumption is normalized. Various schemes have been used, including normalizing to total body weight; fat-free mass; body weight raised to various exponents (for example, 0.75 or 0.67); or the combined weight of various internal organs (for example, heart, liver, kidney and brain; or liver, mesenteric fat and muscle)^{18,19}. Part of the difficulty is that, as mentioned with regard to WAT, the effects of CR are not uniform across all tissues and organs. Given that different tissues have different metabolic requirements, scaling to body mass or some exponent of total body mass may lead to erroneous interpretation of the metabolic data when comparing a normally fed animal to one that has undergone a CR regimen.

Besides an alteration in overall metabolic rate, another bioenergetic effect of CR is a decreased reliance on carbohydrates and an increased utilization of fatty acid oxidation (FAO)⁸. Similarly, certain long-lived mice, such as the previously discussed Ames mouse, also display greater reliance on lipids than carbohydrates when compared to wild-type mice²⁰. Most of these measurements regarding age-dependent changes in substrate utilization are based on the respiratory quotient, a global marker of CO₂ eliminated over consumed oxygen that indirectly reflects the ratio of carbohydrate to fatty acid metabolism. Recent more direct measurements of substrate utilization in aging human muscle demonstrated that there is also evidence of impaired glucose utilization²¹. The results of some randomized human trials indicate that caloric restriction bolsters the body's ability to adapt to use whatever energy substrates are available, be those glucose or fatty acids²². This adaptation is known as metabolic flexibility, a property that has long been linked to metabolic health and, increasingly, longevity⁷. There is also a strong link between CR and reduced inflammation, shown, for instance, in randomized data from nonhuman primates²³. The exact molecular basis for these effects is unclear; however, it is of interest that a recent report has linked a rise in serum β -hydroxybutyrate, a ketone metabolite known to increase during fasting, to inhibition of the inflammasome²⁴. Such metabolic-based inhibition of inflammasome activation may be at least one mechanism through which CR could suppress detrimental activation of the immune system. Other possibilities include a CR-induced delay in thymic involution, thus preserving T cell function²⁵, or a reduction in oxidative stress-induced inflammation²⁶. Finally, there are two ongoing randomized trials involving the effects of CR on healthy rhesus monkeys^{27,28}. These studies differ in their definition of the control and CR diets, and these differences in experimental design most likely explain the studies' conflicting results concerning the effects of CR on nonhuman-primate lifespan: one study has shown an increase in interim lifespan whereas the other has not. However, both groups have seen beneficial effects of CR on certain metabolic parameters, including weight and triglyceride levels.

Nutrient sensing and longevity

mTOR. Numerous studies have attempted to understand how food availability and metabolism are molecularly coupled to lifespan. One clear connection is through the mechanistic target of rapamycin (mTOR). mTOR is a serine-threonine kinase that functions as an intracellular energy sensor and, in mammalian cells, exists in two distinct protein complexes: mTORC1 and mTORC2. In the setting of growth factor stimulation or a rise in intracellular amino acid levels, mTOR is activated and, in turn, regulates a host of downstream events that modulate growth and overall metabolism²⁹. Genetic inhibition

of mTOR signaling can extend lifespan in a wide range of organisms, including worms³⁰, flies³¹ and mice^{32,33}. In yeast and worms, under certain experimental conditions, adding dietary restriction to genetic inhibition of mTOR does not result in further lifespan extension, indicating that mTOR may function as a central mediator of CR's beneficial effects^{34,35}. Although the data suggest that reducing mTOR signaling extends lifespan in a number of species, the exact mechanism through which these effects are achieved remains unclear. This lack of clarity is due to certain key functions of mTOR, which, befitting its role as an energy sensor, regulates a number of downstream effectors that have been linked to lifespan. These roles include the regulation of protein synthesis, mitochondrial function, substrate utilization, insulin signaling and autophagic flux^{29,36}.

Evidence suggests that mTOR can also interact directly with the mitochondria³⁷, although the exact role of mTOR in metabolism seems to be complex and tissue specific. For instance, in transformed mammalian cell lines, there is evidence that mTOR positively regulates mitochondrial respiration^{38,39}, whereas observations in yeast suggest that mTOR acts as a negative regulator of oxygen consumption⁴⁰. This dichotomy is also seen *in vivo*; disruption of mTORC1 activity in mice has been observed to have contrasting effects on mitochondrial function in skeletal muscle and adipose tissue^{41,42}. Interestingly, long-lived mice that are hypomorphic for mTOR expression (i.e., they express mTOR at roughly 25% of wild-type levels) demonstrate no clear alteration in overall metabolic rate³³. Finally, although mTOR acts as an important intracellular energy sensor that is increasingly linked to the beneficial effects of interventions such as CR, it is important to note that this pathway can also act in a non-cell-autonomous fashion. The best example of this may come from analyzing how CR alters intestinal stem cell (ISC) function. Recent evidence suggests that a reduction in mTORC1 activity underlies the beneficial effect of CR on ISC activity in mice. This effect is not mediated by mTOR activity in the stem cell, but rather by the effect of dietary restriction on mTORC1 activity within the surrounding niche cells⁴³.

Sirtuins. The sirtuin family of proteins is another set of potential mediators linking nutrient availability to lifespan. In mammals, there are seven sirtuin family members whose activities include nicotinamide adenine dinucleotide (NAD)-dependent deacetylation of both histone and nonhistone proteins. Increasing evidence suggests that certain sirtuin family members can catalyze the removal of acyl moieties of even longer carbon lengths, including malonyl and succinyl groups, from specific target proteins such as enzymes involved in the tricarboxylic acid cycle (TCA) and fatty acid metabolism⁴⁴. The link between the enzymatic activity of various sirtuin family members with NAD and various TCA intermediates (for example, acetyl CoA, succinyl CoA, etc.) suggests that this family of proteins might sit at the crossroads of metabolism and longevity. In support of such an assumption, initial reports suggested that the sirtuins mediated the longevity effects of CR in both yeast⁴⁵ and flies⁴⁶. However, some latter reports have challenged whether or not sirtuins have a central role in the CR-mediated effects observed in yeast⁴⁷.

In mice, levels of the sirtuin SIRT1 seem to be regulated by nutrient availability^{48,49}, although as discussed for mTOR activity, this might involve tissue-specific regulation⁵⁰. Moreover, mice lacking SIRT1 expression no longer experience an increase in lifespan when placed on a CR regimen⁵¹. Interestingly, brain-specific overexpression of SIRT1 results in long-lived animals which, when they age, seem to have preserved mitochondrial morphology along with increased oxygen consumption and more robust physical activity⁵². Similarly, whole-body

overexpression of SIRT6 can result in lifespan extension for male mice⁵³. The sirtuin family members that localize to the mitochondria (SIRT3, SIRT4, and SIRT5) have widespread metabolic effects, including regulating oxidative phosphorylation⁵⁴, altering fatty acid oxidation⁵⁵ and modulating mitochondrial oxidant production^{56,57}. Proteomic analysis suggests that nearly 65% of mouse mitochondrial proteins are acetylated, and the majority of these have more than one identified acetylation site⁵⁸. Moreover, the mitochondrial acetylome is dramatically reshaped by nutrient availability and by the presence or absence of SIRT3 (ref. 58). Besides increasing lifespan, CR may delay some age-related diseases, and there is evidence that in hearing loss, SIRT3 activity is required to mediate the beneficial effects of CR on this age-related condition⁵⁹. Similarly, the global and presumed beneficial transcriptional program induced by CR requires SIRT3 (ref. 60).

AMPK and autophagy. The metabolic sensor adenosine monophosphate (AMP)-activated protein kinase (AMPK) and the important physiological process of lysosomal-dependent recycling known as autophagy are two other critical links between nutrients and lifespan. When intracellular energy stores are depleted, the levels of lower-energy adenine nucleotides (for example, AMP and adenosine diphosphate (ADP)) increase in proportion to higher-energy forms (ATP). AMPK is activated by binding AMP or ADP, and in turn, AMPK regulates a host of metabolic pathways that, in general, increase energy supplies and reduce energy demand⁶¹. In worms and flies, certain modes of dietary restriction seem to require the activation of AMPK orthologs in order to achieve lifespan extension^{62–64}. However, this requirement often depends on the degree of dietary restriction and the composition of the diet given⁶⁵. Similarly, in mammalian tissues, the activation of AMPK depends on both the tissue and the CR protocol carried out⁶⁶. In mammals, one of AMPK's many targets is UNC-51-like kinase 1 (ULK1), a key regulator of autophagosome formation. The ability of AMPK to regulate autophagy through the phosphorylation of ULK1 (which is the mammalian ortholog of the protein encoded by the yeast essential autophagy-related 1 gene *Atg1*) is increasingly thought to be an important modulator of organismal aging⁶⁷. Interestingly, these effects can function as previously discussed for mTOR signaling in a non-cell-autonomous fashion. For instance, in *Drosophila*, although activation of AMPK in the nervous system triggers induction of autophagy in the brain, it also unexpectedly and simultaneously induces the same effect in distal tissues such as the intestines⁶⁷. These distal effects were attributed to a reduction in circulating insulin-like peptide levels. Autophagic induction seems to be observed in a growing number of pathways that modulate lifespan. For instance, the life-extending effects of mutations in the insulin-like signaling pathway (for example, *Daf-2*) in worms seem to be dependent on a functioning autophagic system⁶⁸. Similarly, the beneficial effects of CR in *Caenorhabditis elegans* or of methionine restriction in yeast do not occur when autophagy is inhibited^{69,70}.

Although evidence suggests that the induction of autophagy is a downstream mediator of various life-extending interventions, there is also evidence that increased autophagy is by itself sufficient to extend lifespan. Increased expression of the essential autophagy-related 5 gene *Atg5* resulted in transgenic mice that had an approximate 17% increase in lifespan when compared to control non-transgenic animals⁷¹. As these mice aged, they also exhibited improvements in insulin sensitivity, redox homeostasis and muscle strength. Similarly, increased expression of the basic helix-loop-helix (HLH) transcription factor HLH-30, the worm ortholog of the mammalian transcription factor EB (TFEB), augments the expression of a number of essential autophagic genes and produces a corresponding increase in lifespan⁷².

IGF and insulin signaling

As suggested in the previous section, there is a strong association, particularly in model organisms, between insulin/IGF-1 signaling and lifespan. This relationship first came to light with analysis of *C. elegans* mutants that lived twice as long as their wild-type controls and that were subsequently shown to have a mutation in a receptor that bears homology to the mammalian insulin/IGF-1 receptor^{73,74}. These Daf-2 mutants have been fundamentally important in dissecting the molecular basis of aging in worms. Subsequent analysis has identified the transcription factor Daf-16 as an important downstream mediator of Daf-2's effects. Daf-16 is homologous to the mammalian FOXO family of forkhead transcription factors, and it was demonstrated to regulate hundreds of downstream targets, including many genes involved in metabolism and the oxidative stress response⁷⁵. A similar reduction of signaling through this axis also results in lifespan extension in *Drosophila*⁷⁶. There is also evidence in mammalian models for the importance of this pathway. We have already discussed the Ames dwarf mouse that has, among other defects, a reduction in insulin/IGF-1 signaling. Similar but more targeted approaches that alter the IGF receptor or IGF-1 bioavailability have led to increased longevity, although sometimes these effects are strain or sex specific^{77–80}. Intriguingly, genetic analyses of long-lived humans have often correlated mutations in the IGF-1 receptor or in FOXO transcription factors with extended health span and longevity^{81,82}. Finally, secreted factors such as insulin/IGF-1, or as mentioned in the context of dwarf mice, secretion of factors such as growth hormone, highlight the role of the somatotrophic axis in coordinating growth, metabolism and longevity⁸³. Similarly, there is growing evidence that neuroendocrine regulation is critically important for both energy homeostasis and aging^{52,84}.

The CR-sensing network. Although we have focused on the various nutrient-sensing pathways in isolation, numerous important connections exist between these pathways (Fig. 2). For instance, in the case of autophagy, both mTOR and AMPK can regulate ULK1 activity⁸⁵. Similarly, the sirtuins, via deacetylation, are also important determinants of autophagic flux⁸⁶. The IGF-insulin signaling pathway, which represents another key axis that connects metabolism with aging, also helps to connect AMPK, mTOR, sirtuins and autophagy together^{48,65}. In many cases, it is difficult to unambiguously determine which component lies upstream or downstream, as for instance, AMPK can be viewed as both a regulator and a target of mTOR signaling. Nonetheless, the tapestry that is emerging is one in which lifespan regulation is woven together by the coordinated activities of various key energy-sensing nodes in the cell.

Mitochondria and aging

As previously mentioned, numerous screens in model organisms have implicated nuclear-encoded mitochondrial proteins in the regulation of lifespan^{3,4}. Most of these genetic interventions altered the expression of components of the electron transport chain that impaired mitochondrial function and yet resulted in a longer lifespan. This is difficult to conceptualize because human aging is generally associated with a decline in mitochondrial function⁸⁷. Similarly, in mouse models that are engineered to markedly increase mitochondrial mutational load, there is evidence of an accelerated aging phenotype^{5,6}. The best explanation for these seemingly conflicting observations is that they vary according to the magnitude of mitochondrial impairment, which, in genetic screens, may have to do with the degree of knockdown achieved. For instance, whereas modest knockdown

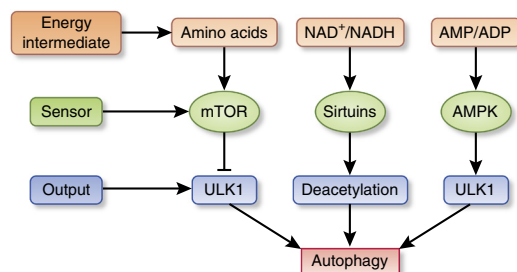


Figure 2 Interaction between various nutrient-sensing and longevity pathways. Three nutrient-sensing pathways are depicted: mTOR, the sirtuin family of proteins and the energy-sensing kinase AMPK. These proteins sense and respond to various different energetic and biosynthetic intermediates. In addition, they have both common and disparate downstream targets, including the serine/threonine-protein kinase ULK1. One area of overlap is in the regulation of autophagy; both the sirtuins and AMPK act to positively regulate autophagy whereas mTOR serves as a negative regulator. For simplicity, interactions between the various sensors (mTOR, sirtuins and AMPK) are not shown. AMP, adenosine monophosphate; ADP, adenosine diphosphate; NAD, nicotinamide adenine dinucleotide.

of various mitochondrial electron transport components leads to increased lifespan in *C. elegans*, higher-efficiency knockdown, which increases mitochondrial impairment, shortens lifespan⁸⁸. There is also genetic evidence in mammalian models that mild impairment of mitochondrial function might extend lifespan. For instance, in a small cohort of animals, mice lacking one allele of the *Coq7* gene (*Mclk1*^{+/-}), which is essential for ubiquinone biosynthesis, lived longer than mice with the *Mclk1*^{+/+} genotype⁸⁹. Interestingly, *Mclk1*^{-/-} mice are not viable, again demonstrating that modest but not severe mitochondrial impairment extends lifespan. A similar genetic alteration that leads to reduced stability for a particular cytochrome *c* oxidase assembly factor also results in a mouse that lived longer than the wild-type control⁹⁰.

Work predominantly in *C. elegans* has demonstrated that a number of pathways are activated after mild mitochondrial impairment⁹¹. The worm ortholog of the mammalian hypoxia inducible factor 1 (HIF-1 α , encoded by *hif-1*) is one important mediator. For several of the long-lived mitochondrial mutants, activation of *hif-1* is required for the observed increased lifespan⁹². Interestingly, the activation of *hif-1* seems to be triggered by an increase in reactive oxygen species (ROS) production resulting from genetically induced mitochondrial impairment. This increase in mitochondrial ROS acts as a signal to activate *hif-1* and thereby increase lifespan. Subsequent studies have demonstrated that *hif-1* not only responds to mitochondrial ROS but also then amplifies ROS production⁹³. As in mammalian cells, this augmented ROS production might have a role in immune surveillance⁹³. Other important mediators of the longevity effects of mild mitochondrial dysfunction in the worm include the p53 ortholog CEP-1 (ref. 94), the homeobox protein CEH-23 (ref. 95), the apoptotic factor CED-3 (ref. 96) and skinhead-1 SKN-1, the *C. elegans* ortholog of the mammalian transcription factor NF-E2 (Nrf2)⁹⁷. Interestingly, in mammalian cells, the activation of Nrf2 is due to a rise in intracellular ROS levels, which triggers a broad transcriptional antioxidant response. Dysregulation of this protective response might be important in mammalian aging⁹⁸. In worms, SKN-1 directly associates with the mitochondria⁹⁹, and this factor mediates the increase in lifespan seen with certain perturbations that simultaneously increase ROS levels and lifespan¹⁰⁰. Similarly, activation of Nrf2 is essential for the anti-carcinogenic effects of caloric restriction¹⁰¹.

Implicit in the previous discussion is the idea that the production of ROS by the mitochondria results in an increase in lifespan. This concept seems antithetical to the long-held belief, first articulated by Denham Harman in the 1950s, that ROS actually fuels the aging process¹⁰². Although there is certainly evidence that mitochondrial ROS can be detrimental and limit lifespan¹⁰³, at least in certain contexts, a small increase in the production of mitochondrial ROS can also trigger the activation of redox-sensitive pathways, resulting in long-lasting protective and life-extending effects. The general concept that a little stress can protect against larger, subsequent stresses is often termed hormesis, and when specifically concerning the mitochondria, mitohormesis^{104–106}. It is also apparent that ROS levels are not the only mediator of mitochondrial stress that can increase lifespan. In one study, it was demonstrated that mitochondrial dysfunction in the brain triggered the production of an as yet undefined secreted factor that could then induce the mitochondrial unfolded protein response (UPR^{mt}) in distal tissues such as the intestines¹⁰⁷. These observations suggest that, at least in the worm, mitochondria in different tissues have mechanisms through which they can communicate, and so expands on the non-cell-autonomous mechanisms already discussed. Similar observations in flies have also demonstrated that mitochondrial dysfunction in skeletal muscle can trigger the UPR^{mt} in that tissue and that it can induce global, non-cell-autonomous metabolic effects¹⁰⁸. These reports highlight the growing interest in the activation of the UPR^{mt} in a, as yet not completely understood, pathway that leads to increased lifespan (Fig. 3). In the presence of certain mitochondrial stresses, various quality-control mechanisms are induced, including unfolded, misfolded or aggregated mitochondrial proteins. This includes activation of the UPR^{mt} mitochondrial-to-nuclear signaling pathway discussed above that ultimately upregulates various mitochondrial chaperone proteins in an attempt to restore protein homeostasis. The intermediaries between the mitochondria and the nucleus may include ROS production but other, non-redox-dependent signaling pathways also seem to be involved¹⁰⁹. In *C. elegans*, one important signaling mediator is the activating transcription factor associated with stress (ATFS-1), which can shuttle between the mitochondria and nucleus¹¹⁰. At present, the mammalian ortholog of ATFS-1 remains unidentified. As discussed above, in the context of increased ROS levels, there are growing indications that the UPR^{mt} not only has a role in aging but also in immune surveillance^{111,112}. Finally, although most studies have been in model organisms, many of these same pathways seem to be operative in mammalian cells, and activation of the UPR^{mt} in mice can increase lifespan¹¹³.

Mitochondrial turnover—namely, the balance between the synthesis of new mitochondria (biogenesis) and the removal of old and damaged mitochondria (mitophagy)—represents an additional process that connects mitochondria to aging. Evidence suggests that interventions that increase longevity, such as CR, stimulate the mitochondrial-biogenesis arm through the induction of peroxisome proliferation-activated receptor gamma, coactivator 1 alpha (PGC-1 α) expression^{114,115}. A decline in biogenesis either through reduced PGC-1 α activity, or via other pathways, is thought to contribute to various age-related pathologies¹¹⁶. Mitochondrial quality control also requires the removal of mitochondria that are old and damaged. Here again, aging seems to produce a gradual decline in autophagic flux¹¹⁷, and it is presumed that the more specialized process of mitophagy follows a similar age-dependent decline. Interestingly, in *C. elegans*, disrupting mitophagy does not alter the lifespan of wild-type worms, but it does abrogate the increased longevity seen in mutants with either mitochondrial dysfunction

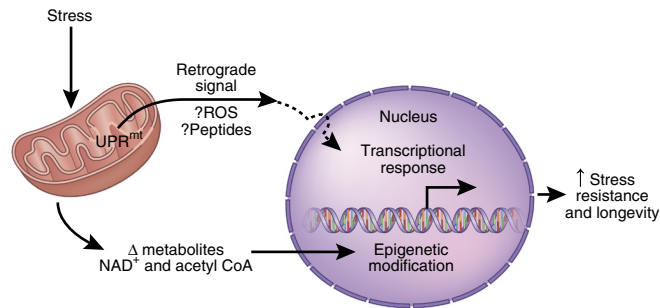


Figure 3 Interaction between the mitochondria and nucleus. Various stresses from outside or inside the mitochondria can trigger activation of the mitochondrial unfolded protein response, UPR^{mt}. The nuclear transcriptional response that follows mitochondrial perturbation most likely involves multiple steps (dotted line) and may be initiated by the release of small amounts of mitochondrial reactive oxygen species (ROS), or in certain cases, the generation of peptides produced (both indicated by question marks) through the actions of mitochondrial proteases. Mitochondrial alterations can also affect intracellular metabolites that are linked to chromatin modification, and this change in the metabolome further connects mitochondrial activity to nuclear gene expression. The net effect of these mitochondrial-nuclear connections is that mitochondrial stresses activate a nuclear program of increased stress resistance. Activation of this mitohormetic pathway is linked to increased longevity.

or reduced intensity of insulin/IGF-1 signaling¹¹⁸. Moreover, this recent study suggested that damaged mitochondria release ROS, which in turn activates SKN-1. The activation of SKN-1 can then simultaneously regulate both the biogenesis and mitophagic pathways in the organism¹¹⁸. Finally, as was seen with the overexpression of core autophagy factors, overexpression of gene products that seem to selectively stimulate mitophagy also seem to extend lifespan¹¹⁹.

Metabolic-based therapy for aging

The identification of key metabolic pathways linked to longevity has provided an impetus for targeting these factors pharmacologically with the hope of delaying aging and ameliorating age-related diseases. One of the first forays into this arena arose from a chemical screen that looked for direct activators of SIRT1. This effort identified the natural plant polyphenol resveratrol as one such sirtuin activator¹²⁰. Considerable controversy has ensued as to whether resveratrol acts as a direct sirtuin activator or whether it instead activates sirtuins in an indirect fashion¹²¹. Although the mechanism is not yet understood, considerable evidence has been obtained demonstrating that resveratrol has beneficial cardiovascular and metabolic effects in animal models^{122,123}, nonhuman primates^{124,125} and human subjects¹²⁶. Yet when rodents are fed a normal diet, resveratrol does not seem to extend their lifespan¹²³. In contrast, newer more specific activators of Sirt1, such as the compound SRT1720, seem to maintain the beneficial metabolic effects that are seen with resveratrol while also providing a notable, albeit modest, increase in lifespan¹²⁷. Similar life-extending benefits have been observed when mice are treated with the mTOR inhibitor rapamycin¹²⁸.

The anti-diabetic drug metformin, which acts at least in part to inhibit mitochondrial function and stimulate AMPK activity, has also shown promise. In worms, metformin extends lifespan by means of a mitohormetic effect involving the production of ROS¹²⁹. Similar data has been obtained in mice from cases in which, again, metformin is able to improve healthspan and lifespan¹³⁰. Befitting a drug that may slow aging, metformin seems to slow the development of age-related diseases. Most notably, epidemiological evidence suggests that

diabetic patients taking metformin have somewhere in the range of a 30–50% decline in expected cancer incidence¹³¹. Given that sirtuins, mTOR and AMPK all regulate autophagy, it is likely that some of the beneficial effects of resveratrol, rapamycin and metformin involve this pathway. Other naturally occurring autophagy activators, such as the polyamine spermidine, can also extend lifespan in various organisms and also seem to slow debilitating conditions such as age-related decline in cognition^{132,133}.

Finally, there is considerable interest in finding strategies to boost intracellular NAD metabolite levels to counteract aging and age-related disease. Biochemical levels of tissue NAD⁺ seem to decline as a function of age in both model organisms and people¹³⁴. Although the basis for this age-dependent decline is incompletely understood, a drop in NAD⁺ metabolite levels seems to reduce sirtuin activity, as these proteins are by and large NAD⁺-dependent enzymes. Augmenting NAD⁺ levels activates the UPR^{mt} pathway, improves metabolic parameters, and remarkably, extends lifespan^{116,135–137}. Other evidence suggests that restoring ‘youthful’ NAD⁺ levels may delay the occurrence of age-dependent diseases¹³⁸.

Conclusions

As discussed, a number of observations in a wide range of experimental models have linked metabolism to aging. Nonetheless, despite considerable progress in the field, at present, aging lacks a clear, unifying theory into which all these disparate observations can be easily incorporated. Unlike our understanding of cancer progression, which is firmly rooted in a genetic model propelled by the gain and loss of oncogenes and tumor suppressors, or cardiovascular disease, which has clearly delineated steps that accelerate the formation of lipid-laden plaques, aging still largely represents a collection of intriguing yet unconnected data sets. Although the free-radical theory of aging held sway for a time¹⁰², it is clear that the life-extending benefits of low-level ROS, as described above, cannot be readily incorporated into such a theory. How then can we frame the relationship between metabolism, energetics and aging? In my mind, the answer may revolve around the use and partition of energy by an organism.

Since the 1920s, when Raymond Pearl described his rate-of-living hypothesis, a relationship between lifespan and metabolic rate has been postulated. Yet it is also clear that metabolic rate alone is not an accurate predictor of lifespan. For instance, as discussed above, CR extends lifespan but has no major effect on adjusted metabolic rate. In addition, although there is a linear relationship between metabolic rate and lifespan for many organisms, various animals (for example, naked mole rats, bats, humans, etc.) clearly fall off the curve¹³⁹. The energy used by an organism needs to be partitioned into various tasks, including normal daily activities, such as movement and feeding, the synthesis of new molecules and the maintenance of existing structures. The latter reflects the energy required to repair damaged membranes, proteins and nucleic acids. It may be instructive to view interventions such as CR, or genetic manipulations such as reducing mTOR activity, as metabolic strategies to reset this partition of energy (Fig. 4). In such a scenario, these life-extending interventions result in a larger proportion (and absolute amount) of resources going toward maintenance functions and away from new growth and biosynthetic activities. In such an energy-partition theory of aging, this reset could be triggered by intracellular stress-signaling pathways as described but would also require coordination throughout the organism, presumably through non-cell-autonomous means. The existence of these circulating factors is hinted, as discussed,

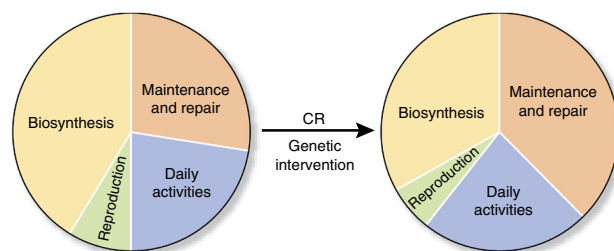


Figure 4 Energy partition and lifespan. Shown is a conceptual framework to potentially explain how environmental or genetic interventions that extend lifespan may alter how the organism uses energy. In general, life-extending interventions shift energy away from reproductive and biosynthetic processes and toward processes involving maintenance and repair. In such a scenario, long-lived animals might have an overall metabolic rate that is unchanged or only modestly reduced. However, the partitioning of energy is shifted away from growth and reproduction and toward energy-consuming processes that are devoted to repairing and maintaining existing structures. CR, caloric restriction.

by various observations in model organisms. In higher organisms, hormones such as fibroblast growth factor 21 (FGF21) are secreted in response to stresses such as prolonged fasting or mitochondrial dysfunction¹⁴⁰. Interestingly, FGF21 is known to inhibit somatic growth, and when overexpressed, to extend the lifespan of mice¹⁴¹. The elucidation of additional circulating factors, as well as intracellular mediators that reset metabolism and potentially shift the energetic balance toward reparative processes, may provide a wealth of new therapeutics for aging and age-related diseases. Perhaps then we may finally understand the complex relationship between metabolism and aging. We might also, for the first time, fully understand the simple advice that Luigi Cornaro prescribed 500 years ago.

ACKNOWLEDGMENTS

I am grateful to members of my laboratory for their helpful suggestions, and in particular to I. Rovira for help with the manuscript. This work is supported by funds from the US National Institutes of Health Intramural Program and from The Leducq Foundation.

COMPETING FINANCIAL INTERESTS

The author declares no competing financial interests.

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